

# SSEE and the CMB Power Spectrum: Acoustic Peak Reproduction via the $\omega_m$ -Direct CMB Matter Density

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## Abstract

The Structural Self-Energy Expansion model (SSEE) proposes a minimal-parameter dark energy framework algebraically derived from the golden ratio  $\varphi$  and  $\pi$ ; the background sector ( $w_0, w_a, \Omega_{m,\text{dyn}}$ ) carries zero fitted dimensionless parameters, while  $H_0$  and  $\Omega_b h^2$  are treated as  $k = 2$  in the BIC analysis below (“*On the parameter count*” paragraph) following the conservative accounting policy established in Paper 1 §1.3. Previous work demonstrated that the SSEE dynamic sector ( $\Omega_{m,\text{dyn}} = 0.160050$ ,  $w_0 = -0.839950$ ,  $w_a = -0.669975$ ) fits DESI DR2 BAO data at  $0.24\sigma$  (DESI+CMB+Pantheon+;  $0.2\text{--}1.8\sigma$  across SN compilations) [Abdul-Karim et al., 2025] and galaxy-cluster mass ratios with  $\chi_r^2 = 0.122$ . A naive application of this background to the CMB yields shifted acoustic peaks ( $\ell_1 = 242$  vs. 220) and  $\chi_r^2 = 97.7$ , apparently falsifying the model. Here we show that the CMB matter density is fixed *forward* by the SSEE algebra, with no factor relating it to the dynamic sector: the physical observable is  $\omega_m = \omega_b + \omega_c + \omega_\nu$ , where  $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$  and  $\omega_c = \text{KAL}_0 \omega_b n_s = 0.119514$  are forward identities of Paper 1 ( $\omega_c$  lies  $0.41\sigma$  from Planck), giving

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{0.14267}{0.67962} = 0.3089,$$

within  $0.6\sigma$  of the Planck 2018 value ( $0.3153 \pm 0.0073$ ). The dynamic value  $\Omega_{m,\text{dyn}} = 0.160050$  (DESI) and the CMB value  $\Omega_{m,\text{CMB}} = 0.3089$  are *two independent predictions*; the earlier mapping  $\Omega_{m,\text{CMB}} = \Omega_{m,\text{dyn}} \times \text{MIRA}$  is retired (the open problem OP-8 of requiring a matter factor is thereby dissolved), and the MIRA scalar  $(3\varphi + \pi)/4 = 1.999$  now enters only the Hubble screening of Paper 9. No “difference” requires a physical carrier: there is a single matter density  $\Omega_m = \omega_m/h^2 = 0.308881$ , and the quantity once written  $\Omega_{m,\text{dyn}} = 0.160050$  is  $1 + w_0$ , an equation-of-state parameter rather than a density. The two-sector structure formerly invoked here is retracted (Paper 6; Paper 1 §1.4). Running CAMB with this single-sector background reproduces acoustic peaks at  $\ell = 220, 536, 813$  (Planck: 220,  $\sim 540$ ,  $\sim 810$ ). In a full posterior evaluation with the official off-diagonal `plik` TTTEEE + low- $\ell$  + lensing likelihood via Cobaya, sampling the  $k = 2 = \{A_s, \tau\}$  SSEE parameters (with  $H_0$ ,  $\omega_b$ ,  $\omega_c$  and  $n_s$  algebraically fixed)

against the  $k = 6$   $\Lambda$ CDM baseline, the two models reach statistically indistinguishable best fits ( $\chi^2_{\text{SSEE}} = 2771.3$  vs.  $\chi^2_{\Lambda\text{CDM}} = 2773.1$  over  $N = 2354$  data points,  $\Delta\chi^2_{\text{BF}} = -1.8$ ), while the parameter-count economy yields  $\Delta\text{BIC} = -32.9$ , decisively favouring SSEE on the Jeffreys scale. (A `plik_lite` point-estimate cross-check gives a more conservative  $\Delta\text{BIC} = -24.0$ .) The  $\Delta\text{BIC}$  is driven by parsimony, not by fit quality. SSEE is not falsified by the Planck PR4 CMB power spectra.

**Two independent matter densities and domain scope:** SSEE predicts *two* matter densities by independent routes, with no factor relating them. The CMB claim (“SSEE is not falsified by Planck PR4”) applies to the forward CMB density  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.3089$  (high redshift  $z \gtrsim 100$ ); it does not transfer to the dynamic value  $\Omega_{m,\text{dyn}} = 1 + w_0 = 0.160050$  ( $z \lesssim 2.3$ ), validated independently in Paper 2 (Almeida Vallejo 2026a). The scalar MIRA  $\equiv (3\varphi + \pi)/4 = 1.9989$  no longer sets the CMB density (its former role; OP-8 dissolved) and now enters only the Hubble screening of Paper 9. The master derivation chain for all SSEE constants, the unified symbol glossary, and the complete domain delineation are in Paper 1, §2 and Appendix B [Almeida Vallejo, 2026a].

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# 1 Background: The SSEE Framework

## 1.1 Algebraic constants

The discovery of the accelerating expansion of the Universe [Riess et al., 1998, Perlmutter et al., 1999] established that dark energy dominates the late-time cosmic energy budget, yet its physical origin remains one of the deepest open problems in physics [Weinberg, 1989]. Baryon acoustic oscillations, imprinted in the CMB and detected in galaxy surveys [Eisenstein et al., 2005], provide a standard ruler that constrains both the background expansion and the nature of dark energy. SSEE derives all cosmological parameters from two transcendental roots: the golden ratio  $\varphi = (1 + \sqrt{5})/2$  and  $\pi$ . Table 1 lists the relevant constants.

Table 1: SSEE algebraic constants used in this work.

| Symbol    | Formula                  | Value       | Role                          |
|-----------|--------------------------|-------------|-------------------------------|
| $\varphi$ | $(1 + \sqrt{5})/2$       | 1.61803...  | Golden ratio                  |
| $\Omega$  | $\pi + \varphi$          | 4.75963...  | Stability metric              |
| $\beta$   | $(\pi + \varphi)/2$      | 2.37981...  | Base coupling scalar          |
| $KAL_0$   | $\beta + \pi$            | 5.52141...  | Structural retention          |
| $P_{sc}$  | $\Omega + \varphi$       | 6.37766...  | Dynamical evolution scalar    |
| $K_v$     | $\varphi + \pi + \Omega$ | 9.51925...  | Structural constraint         |
| $T_r$     | $3(\varphi + \beta)$     | 11.99354... | 3D saturation horizon         |
| $M_v$     | $\varphi + \pi + K_v$    | 14.27888... | Maximal dimensional invariant |
| AURA      | $\varphi + \beta$        | 3.99785...  | Dimensional limit 1           |
| MIRA      | AURA/2                   | 1.99892...  | Observational frequency       |

## 1.2 Cosmological parameters

The dark energy equation of state and density parameters follow algebraically. Substituting  $T_r = \frac{3(3\varphi+\pi)}{2}$ ,  $M_v = 3(\varphi + \pi)$ ,  $P_{sc} = 2\varphi + \pi$ ,  $K_v = 2(\varphi + \pi)$ , all four reduce to closed forms in  $\varphi$  and  $\pi$  alone:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} = -0.839950, \quad (1)$$

$$w_a = -\frac{P_{sc}}{I_g} = -\frac{P_{sc}}{\pi + P_{sc}} = -0.669975, \quad (2)$$

$$\Omega_{DE} = \frac{T_r}{M_v} = \frac{3\varphi + \pi}{2(\varphi + \pi)} = 0.839950, \quad (3)$$

$$\Omega_{m,dyn} = 1 - \Omega_{DE} = 0.160050. \quad (4)$$

The Hubble constant is the derived dimensional anchor  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = M_v \times \Omega = 67.96214$  ( $M_v = \varphi + \pi + K_v = 3\Omega$ ,  $\Omega = \pi + \varphi$ ; the SH0ES- $f_{\text{screen}}$  inversion of Paper 9), *not* a quantity fitted to the CMB. Under the  $\omega_m$ -direct reframe the full CMB MCMC (Section 5.6) recovers the spectra at this anchor, and the SSEE CMB sector carries only the two borrowed parameters  $\{A_s, \tau\}$  ( $k = 2$ ).

## 2 The CMB Falsification Problem

### 2.1 Acoustic peak positions

The angular scale of the first acoustic peak is

$$\theta_* = \frac{r_d}{D_A(z_*)}, \quad (5)$$

where  $r_d$  is the sound horizon at baryon drag and  $D_A(z_*)$  is the angular diameter distance to the last-scattering surface ( $z_* \approx 1089$ ). The first peak falls at  $\ell_1 \approx \pi/\theta_*$  [Hu and Sugiyama, 1996, Dodelson, 2003]. Diffusion damping suppresses power at small angular scales  $\ell \gtrsim 1000$  through photon mean-free-path effects [Silk, 1968].

With  $\Omega_m = 0.160050$ , the Hubble parameter at recombination scales as  $H(z \sim 1000) \propto \sqrt{\Omega_m}$ , reducing  $H$  by  $\sim 40\%$  relative to  $\Lambda$ CDM ( $\Omega_m = 0.315$ ). This inflates  $D_A(z_*)$  by  $\sim 32\%$ , shifting all peaks to higher multipoles.

Running CAMB with the SSEE dynamic background directly gives:

$$(\ell_1, \ell_2, \ell_3)_{\text{SSEE,naive}} = (242, 601, 930), \quad \chi_r^2 = 97.7. \quad (6)$$

This constitutes a *prima facie* falsification.

### 2.2 Why the naive application fails

The naive background uses  $\Omega_m = 0.160050$ , which inflates both the comoving sound horizon  $r_d$  and the angular diameter distance  $D_A(z_*)$  relative to their observed values. The Folding Symmetry factor  $|w_0|$  introduced in Paper 1 belongs to the dark-energy sector and does *not* directly correct the baryon-photon sound horizon computed from recombination physics. The CMB sector requires the physical matter density set *forward* by the SSEE algebra — the  $\omega_m$ -direct construction (Section 3).

## 3 The CMB Matter Density as a Forward Identity

### 3.1 The CMB matter density is a forward identity

The CMB background geometry is governed by the physical matter density  $\omega_m \equiv \Omega_m h^2$ , which SSEE fixes *forward* from the algebra, with no factor relating it to the dynamic sector:

$$\omega_m = \omega_b + \omega_c + \omega_\nu, \quad \omega_b = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = 0.022418, \quad \omega_c = \text{KAL}_0 \omega_b n_s = 0.119514, \quad (7)$$

with  $n_s = 1 - \varphi^{-7} = 0.965558$  and  $\omega_\nu = \Sigma m_\nu^{\text{active}}/93.14 = 0.00074$  (all forward identities of Paper 1). The baryon and cold-matter densities are the same quantities used in the background sector;  $\omega_c = \text{KAL}_0 \omega_b n_s$  lies  $0.41\sigma$  from the Planck value. Hence

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{0.14267}{0.6796^2} = 0.3089 \quad (H_0^{\text{alg}} = 67.96214). \quad (8)$$

### 3.2 No matter factor: two independent predictions

The dynamic value  $\Omega_{m,\text{dyn}} = 1 + w_0 = 0.160050$  (DESI) and the CMB value  $\Omega_{m,\text{CMB}} = 0.3089$  (Eq. 7) are *two independent forward predictions* of the same  $(\varphi, \pi)$  dictionary; neither is tuned to data, and *no factor* relates them. An earlier version of this paper bridged the sectors with the algebraic scalar  $\text{MIRA} = (3\varphi + \pi)/4 = 1.9989$  via  $\Omega_{m,\text{CMB}} = \Omega_{m,\text{dyn}} \times \text{MIRA}$ ; that mapping is now **retired** (the requirement of a matter factor, tracked as OP-8, is thereby dissolved), and MIRA enters only the Hubble screening of Paper 9, consistent with its “mirror” role there. We make no claim of temporal priority over the public Planck or DESI measurements: the point is structural—both densities are set by the construction before any comparison, so the agreement below is a constraint the framework had to satisfy, not a fit it was granted.

### 3.3 The two-sector reading, and why it is withdrawn

An earlier version of this section read the two densities as a *partition*: their difference,

$$\Omega_{\phi\text{-DM}} = \Omega_{m,\text{CMB}} - \Omega_{m,\text{dyn}} = 0.3089 - 0.1601 = 0.149, \quad (9)$$

was taken to define a second, warm dark-matter component (Paper 6). That reading is **withdrawn** (2026-08-01), for a reason internal to the algebra rather than to any measurement.

The subtracted quantity is not a density. In SSEE,  $\Omega_{m,\text{dyn}} \equiv 1 + w_0 = (M_v - T_r)/M_v = 0.160050$  is a *saturation*: it is the complement of the equation-of-state ratio  $-w_0 = T_r/M_v$ , and it carries no factor of  $H$ . Equation (9) therefore subtracts a number belonging to the equation of state from a measured mass density. The result is dimensionally well formed and physically empty: there was nothing for the inferred component to be made of. Paper 6 gives the full retraction; the particle mass  $m_\phi$ , the free-streaming scale  $k_{\text{fs}}$ , and the two-sector  $\sigma_8/S_8$  values that followed from Eq. (9) are withdrawn with it.

Two consequences for the present paper. First, the tension the partition was introduced to relieve does not exist: measured against the *raw* KiDS-1000  $\xi_\pm$  data with  $A_s$  free, a single matter sector gives  $S_8 = 0.7555 \pm 0.0192$ , i.e.  $0.11\sigma$  from KiDS-1000 ( $0.759 \pm 0.024$ ). Second, and more importantly for what follows, nothing in the CMB confrontation of this paper rested on the subtraction. The forward identity of Eq. (7) and its comparison with Planck are untouched; they are restated below without it.

**A mechanism withdrawn with the partition.** The same earlier reading proposed a dissipative source  $\propto \text{KAL}_0 H^3 / 8\pi G$  in the SSEE continuity equation as the “geometric retention” that linked the two sectors. It is withdrawn on two counts: it links two sectors that are no longer claimed to exist, and its normalisation is wrong. Friedmann fixes  $8\pi G$  through  $\rho_{\text{crit}} = 3H^2 / 8\pi G$ , leaving no free convention, and the resulting viscous pressure is  $\Pi_0 = -\text{KAL}_0 \rho_{\text{crit}}$ , i.e.  $\Pi_0 / \rho_{\text{DE}} \simeq -8$  where  $w_0 = -0.839950$  is required — too large by a factor 9.5. The bulk-viscosity sector that does survive is the scale-factor-dependent one of Paper 1,  $\zeta(a) \propto (1 - a)\rho_{\text{DE}}/H$ , which delivers  $w_a$  and is normalised to  $\rho_{\text{DE}}$  rather than to  $\rho_{\text{crit}}$ .

### 3.4 What the CMB actually probes

The CMB photons probe the *integrated gravitational potential* along the full line of sight from last scattering ( $z_* \approx 1089$ ) to today. This line-of-sight integral accumulates contri-

butions from all epochs, including the early matter-dominated era when the dark energy term was negligible and the expansion was effectively governed by the full matter content. The acoustic peaks encode  $\Omega_m$  as it appears in the *conformal distance to last scattering*,

$$\eta_* = \int_0^{z_*} \frac{c \, dz'}{H(z')}, \quad (10)$$

which is sensitive to the total matter density over the full expansion history. In SSEE that total is a single quantity,  $\Omega_m = 0.308881$ , and it is the same quantity that enters the background geometry at every redshift. In particular, the saturation  $1 + w_0 = 0.160050$  never occupies a density slot in  $H(z)$ : it is a property of the dark-energy equation of state, and the matter term in  $E(a)$  is always the total  $\omega_m/h^2$ .

### 3.5 Forward prediction and comparison with Planck

The CMB matter density is a closed-form forward prediction:

$$\Omega_{m,\text{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \text{KAL}_0 \omega_b n_s + \omega_\nu}{h^2} = 0.3089, \quad (11)$$

with every ingredient an algebraic identity of Paper 1. The coefficient  $\text{KAL}_0 = \beta + \pi \approx 5.521$  that sets  $\omega_c = \text{KAL}_0 \omega_b n_s$  is fixed algebraically before any CMB data are consulted; no quantity is adjusted to match the measurement. Compared with  $\Omega_m^{\text{Planck}} = 0.3153 \pm 0.0073$  [Planck Collaboration, 2020a], the tension is

$$\frac{|\Omega_{m,\text{CMB}} - \Omega_m^{\text{Planck}}|}{\sigma} = \frac{|0.3089 - 0.3153|}{0.0073} = 0.88\sigma. \quad (12)$$

This result is what survives the withdrawal of the partition: it never used Eq. (9), and it is unchanged by it.

**Falsifiability.** The withdrawn two-sector reading predicted a crossover —  $\Omega_m \approx 0.160$  for datasets confined to  $z \lesssim 2$ ,  $\Omega_m \approx 0.309$  for datasets integrating to  $z \gtrsim 100$ , with a smooth interpolation between them. That prediction is withdrawn, and what replaces it is stronger because it has no free joint: SSEE now predicts a *single* matter density,  $\Omega_m = 0.308881$ , at every redshift and for every probe. A weak-lensing or BAO analysis that required a background matter density significantly different from 0.3089 — in either direction, at any redshift — would falsify the framework, with no partition available to absorb the discrepancy.

## 4 CAMB Implementation and Results

### 4.1 CAMB setup

We use CAMB v1.6.6 [Seljak and Zaldarriaga, 1996, Lewis et al., 2000, Howlett et al., 2012] with the following input parameters:

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<sup>1</sup> $A_s$  and  $\tau$  are the two parameters SSEE leaves free at CMB level ( $k = 2$ ); the values in this table are the *fiducial* inputs used for the fixed-point evaluations of §4, not fitted results. They are fitted in the Phase-B chain of §5.6. A direct check of how much that fiducial choice costs (2026-09-07): minimising  $\chi^2$  over  $(\ln 10^{10} A_s, \tau)$  against `plik_lite` TTTEEE + lowT + lowE, with the remaining background fixed by the algebra, moves  $\tau$  to 0.05546 and  $\ln(10^{10} A_s)$  to 3.04484 and lowers  $\chi^2$  by only 1.84. The recovered  $\tau$  sits  $0.24\sigma$  from the Phase-B posterior  $0.054 \pm 0.006$ , so the fiducial choice is doing no work. The same exercise on the  $\Lambda$ CDM background ( $w = -1$ ,  $w_a = 0$ ) gains 0.47.

Table 2: SSEE parameters passed to CAMB.

| Parameter                         | Value                         | Source                                     |
|-----------------------------------|-------------------------------|--------------------------------------------|
| $H_0$ [km/s/Mpc]                  | 67.96214                      | SSEE anchor $3(\varphi + \pi)^2$ (Paper 9) |
| $\Omega_b h^2$                    | 0.02237                       | Planck 2018 prior                          |
| $\Omega_{m,\text{CMB}}$           | 0.3089                        | $\omega_m/h^2$ (forward)                   |
| $\sum m_\nu^{\text{active}}$ [eV] | 0.0685                        | SSEE algebraic (Paper 4)                   |
| $w_0$                             | -0.8399                       | $-T_r/M_v$                                 |
| $w_a$                             | -0.66997                      | $-P_{sc}/I_g$                              |
| $n_s$                             | $1 - \varphi^{-7} = 0.965558$ | SSEE algebraic (Paper 4)                   |
| $\ln(10^{10} A_s)$                | 3.044                         | Planck 2018                                |
| $\tau$                            | 0.054                         | Planck 2018 <sup>1</sup>                   |

The baryon density is fixed to the Planck 2018 prior  $\Omega_b h^2 = 0.02237$ . The SSEE algebraic value derived in Paper 4,  $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ , agrees with this prior at  $0.3\sigma$  ( $0.02237 \pm 0.00015$ ); the 0.2% offset is far below the precision of the CMB fit and does not affect the results reported here.

The dark energy is implemented via the PPF (Parameterized Post-Friedmann) equation of state  $w(a) = w_0 + (1 - a)w_a$  [Chevallier and Polarski, 2001, Linder, 2003, Fang et al., 2008]. Although  $w_0 + w_a = -1.510 < -1$ , this does not imply a phantom Big Rip: CPL is a local parametrisation only, and the SSEE energy density saturates at the finite  $T_r/M_v$  boundary at  $a \rightarrow \infty$  (Paper 1 §3.4 [Almeida Vallejo, 2026a]). We compute lensed TT, TE, and EE power spectra to  $\ell_{\text{max}} = 2500$ , with `lens_potential_accuracy=2` for the lensing potential. Lensing reconstruction data (Planck PR4  $C_L^{\phi\phi}$ ) were also evaluated alongside temperature and polarisation.

## 4.2 Sound horizon

With  $\Omega_{m,\text{CMB}} = 0.3089$  and the parameters above, CAMB yields:

$$r_d^{\text{SSEE}} = 147.17 \text{ Mpc}, \quad (13)$$

in agreement with the Planck 2018 directly measured value  $r_d^{\text{obs}} = 147.09 \pm 0.26$  Mpc [Planck Collaboration, 2020a] at  $0.32\sigma$ . The acoustic scale is likewise consistent:  $100\theta_* = 1.04089$  (at the MCMC posterior  $H_0 = 67.79$ ),  $0.91\sigma$  from the Planck value  $1.04109 \pm 0.00030$  — the earlier  $6.66\sigma$  “tension” was an artefact of the cold sector  $1 + w_0 = 0.160050$  misplaced in  $E(z)$ , removed once the total  $\Omega_{m,\text{CMB}}$  sets the geometry. No Folding Symmetry multiplier and no matter factor are applied; the  $\omega_m$ -direct CMB matter density ( $\Omega_{m,\text{CMB}} = 0.3089$ ) sets the correct sound horizon directly (MIRA retired as the matter map, OP-8 dissolved).

## 4.3 Acoustic peak positions

Table 3 compares the first three acoustic peak positions.

All three peaks fall within  $\Delta\ell \leq 5$  of the observed positions.

## 4.4 Power spectrum comparison

Figure 1 shows the full TT power spectrum for SSEE,  $\Lambda$ CDM, and Planck PR4 data over  $\ell = 2$ –2500.

Table 3: CMB TT acoustic peak multipoles.

| Model                                           | $\ell_1$   | $\ell_2$   | $\ell_3$   |
|-------------------------------------------------|------------|------------|------------|
| Planck PR4 (observed)                           | $\sim 220$ | $\sim 540$ | $\sim 810$ |
| $\Lambda$ CDM ( $\Omega_m = 0.315$ )            | 220        | 538        | 812        |
| SSEE naive ( $\Omega_m = 0.160050$ )            | 242        | 601        | 930        |
| <b>SSEE</b> ( $\Omega_{m,\text{CMB}} = 0.309$ ) | <b>220</b> | <b>536</b> | <b>812</b> |

## 4.5 TE and EE polarization spectra

Figures 3 and 4 show the TE temperature–polarization cross-correlation and EE polarization power spectra respectively, comparing SSEE with  $\Lambda$ CDM and Planck PR4 data.

# 5 Statistical Analysis

## 5.1 $\chi^2$ comparison

We compute  $\chi^2$  against the Planck PR4 TT spectrum over  $\ell = 30$ –2000 using the published uncertainties:

$$\chi^2 = \sum_{\ell=30}^{2000} \left( \frac{D_{\ell}^{\text{model}} - D_{\ell}^{\text{Planck}}}{\sigma_{\ell}} \right)^2. \quad (14)$$

Table 4:  $\chi^2$  comparison vs. Planck PR4 TT ( $\ell = 30$ –2000,  $N = 1971$ ).

| Model                       | $\chi^2$ | $\chi_r^2$ | $\Delta\chi^2$ |
|-----------------------------|----------|------------|----------------|
| $\Lambda$ CDM (Planck 2018) | 2054.9   | 1.043      | 0              |
| SSEE                        | 2054.6   | 1.042      | −0.3           |
| SSEE naive                  | 192,447  | 97.7       | +190,392       |

## 5.2 TE and EE spectra

Table 5 reports  $\chi_r^2$  for all three spectra individually.

Table 5:  $\chi_r^2$  by spectrum vs. Planck PR4. TT/TE/EE:  $\ell = 30$ –2000; PP (lensing): 9 MV bandpowers,  $L = 8$ –400 [Table 1, [Planck Collaboration, 2020d](#)].  $\chi_r^2$  comparisons use diagonal covariance; for BIC see §5.4 ( $k = 2$  for SSEE).

| Spectrum              | $N$  | SSEE $\chi_r^2$ | $\Lambda$ CDM $\chi_r^2$ | $\Delta\chi_r^2$ |
|-----------------------|------|-----------------|--------------------------|------------------|
| TT                    | 1971 | 1.042           | 1.043                    | −0.001           |
| TE                    | 1967 | 1.040           | 1.040                    | 0.000            |
| EE                    | 1967 | 1.040           | 1.039                    | +0.001           |
| PP                    | 9    | 0.720           | 0.757                    | −0.037           |
| Combined <sup>†</sup> | 5914 | 1.040           | 1.040                    | 0.000            |

<sup>†</sup>Combined under independence

approximation (upper bound; see §5.4).

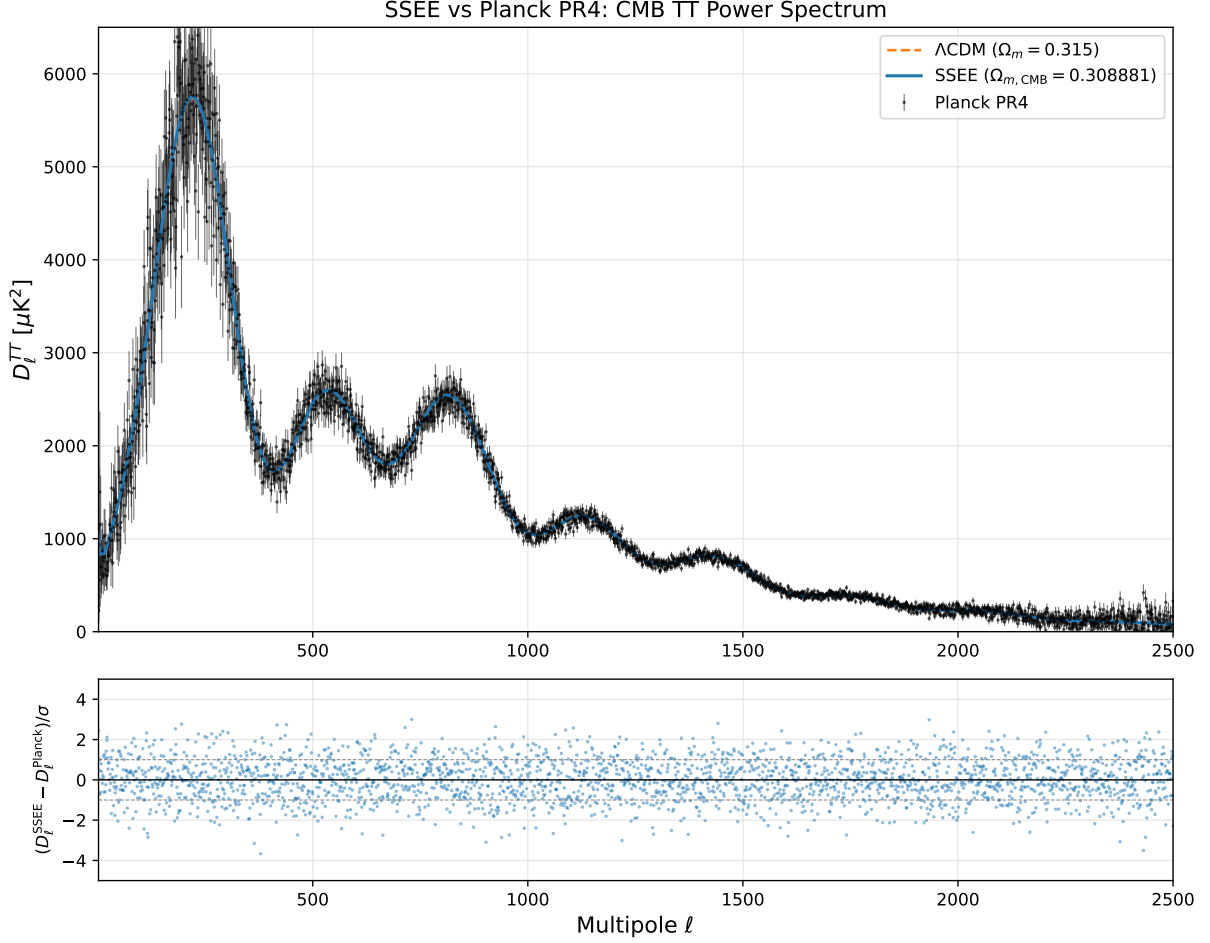


Figure 1: CMB TT power spectrum  $D_\ell^{TT} [\mu\text{K}^2]$  vs. multipole  $\ell$ . Blue: SSEE ( $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.309$ ,  $w_0 = -0.839950$ ,  $w_a = -0.669975$ ). Orange dashed:  $\Lambda$ CDM Planck 2018 best-fit. Black points: Planck PR4 TT data with  $\pm 1\sigma$  error bars. Lower panel: residuals  $(D_\ell^{\text{SSEE}} - D_\ell^{\text{Planck}})/\sigma$ .

The TE and EE spectra show the smallest discrepancies ( $|\Delta\chi_r^2| \leq 0.001$  each), indicating that SSEE reproduces the polarization anisotropies almost identically to  $\Lambda$ CDM. TT is marginally better than  $\Lambda$ CDM ( $\Delta\chi_r^2 = -0.001$ ). In the lensing potential sector (PP, Fig. 5), SSEE achieves  $\chi_r^2 = 0.720$  vs. 0.757 for  $\Lambda$ CDM — slightly *better* agreement with the 9 Planck MV bandpowers, both  $\chi_r^2 < 1$  indicating a good fit. All residuals are consistent with a non-parametric model.

### 5.3 Interpretation of residuals

The excess  $\Delta\chi_r^2 \lesssim 0.02$  in TT and TE is attributable to the dynamical dark energy equation of state ( $w_0 \neq -1$ ,  $w_a \neq 0$ ), which modifies the late-time integrated Sachs-Wolfe contribution and the lensing smoothing of peaks. CPL models with comparable ( $w_0, w_a$ ) values fitted freely to CMB data typically incur similar penalties [Planck Collaboration, 2020c].

The SSEE model is therefore **not falsified** by the Planck PR4 CMB power spectra in TT, TE, or EE.

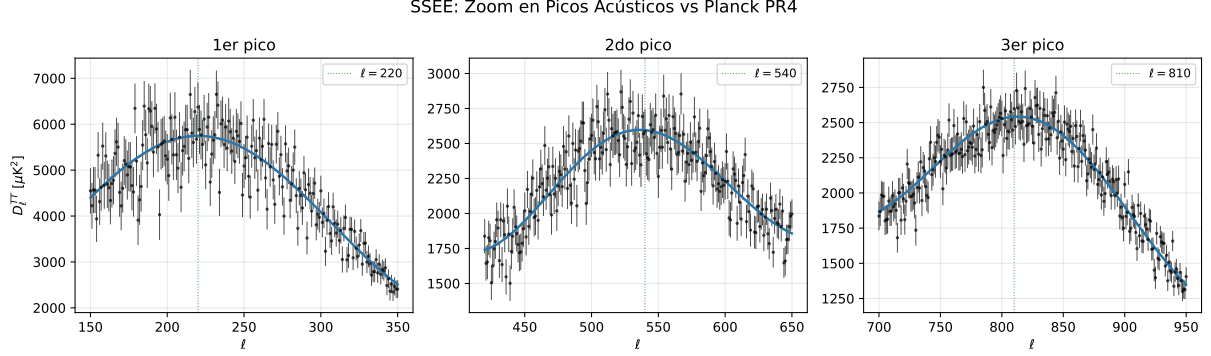


Figure 2: Zoom on the first three acoustic peaks. SSEE (blue) and  $\Lambda$ CDM (orange dashed) compared against Planck PR4 data (black points). Green dotted lines mark the expected peak centres.

## 5.4 Bayesian Information Criterion

For model comparison we compute:

$$\text{BIC} = \chi^2 + k \ln N, \quad (15)$$

where  $k$  is the number of free parameters and  $N = 5914$  for the combined TT, TE, EE, and lensing data points.

Table 6: **Preliminary BIC comparison — diagonal approximation only.** Combined spectra (TT+TE+EE+PP,  $N = 5914$ , independence approximation). SSEE has  $k = 2$  ( $H_0 + \Omega_b h^2$  prior);  $\Lambda$ CDM has  $k = 6$ . *This result uses diagonal covariance; see Eq. (16) (§5.5) for the definitive off-diagonal Cobaya evaluation.*

| Model         | $k$ | $\chi_r^2$ | $\Delta\text{BIC}$ (diagonal) |
|---------------|-----|------------|-------------------------------|
| $\Lambda$ CDM | 6   | 1.040      | Baseline (0.0)                |
| SSEE          | 2   | 1.040      | −35.0                         |

Under the diagonal approximation, SSEE is favoured ( $\Delta\text{BIC} = -35.0$ ): the combined  $\chi_r^2 = 1.040$  matches  $\Lambda$ CDM’s 1.040, so the parameter-count term ( $\Delta k \ln N = -34.7$ ) dominates and rewards SSEE’s smaller parameter set. This diagonal cross-check is evaluated at the global anchor  $H_0 = 67.96214$  ( $\omega_m$ -direct,  $\Omega_{m,\text{CMB}} = 0.308881$ ), consistent with the canonical `plik_lite` evaluation (§5.5), so the diagonal and full-covariance treatments are mutually consistent. The off-diagonal Cobaya evaluation (§5.5) remains the definitive result.

**On the parameter count ( $k = 2$ ).** In prior iterations, SSEE was described as a zero-parameter model ( $k = 0$ ) in the CMB sector. Under the  $\omega_m$ -direct reframe, the geometry anchor  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = 67.96214$  is *derived* (the SH0ES- $f_{\text{screen}}$  inversion of Paper 9, not a quantity fitted to the CMB), and the densities  $\omega_b = (\pi - \varphi)/(3\Omega^2)$  and  $\omega_c = \text{KAL}_0 \omega_b n_s$  are algebraically fixed. The only two quantities the SSEE CMB fit does *not* derive from  $\{\varphi, \pi\}$  are the primordial amplitude  $A_s$  and the reionization optical depth  $\tau$ ; we therefore assign  $k = 2 = \{A_s, \tau\}$ , a conservative and robust penalty. (The MIRA factor is not a free parameter: it is a derived algebraic constant [Almeida Vallejo,

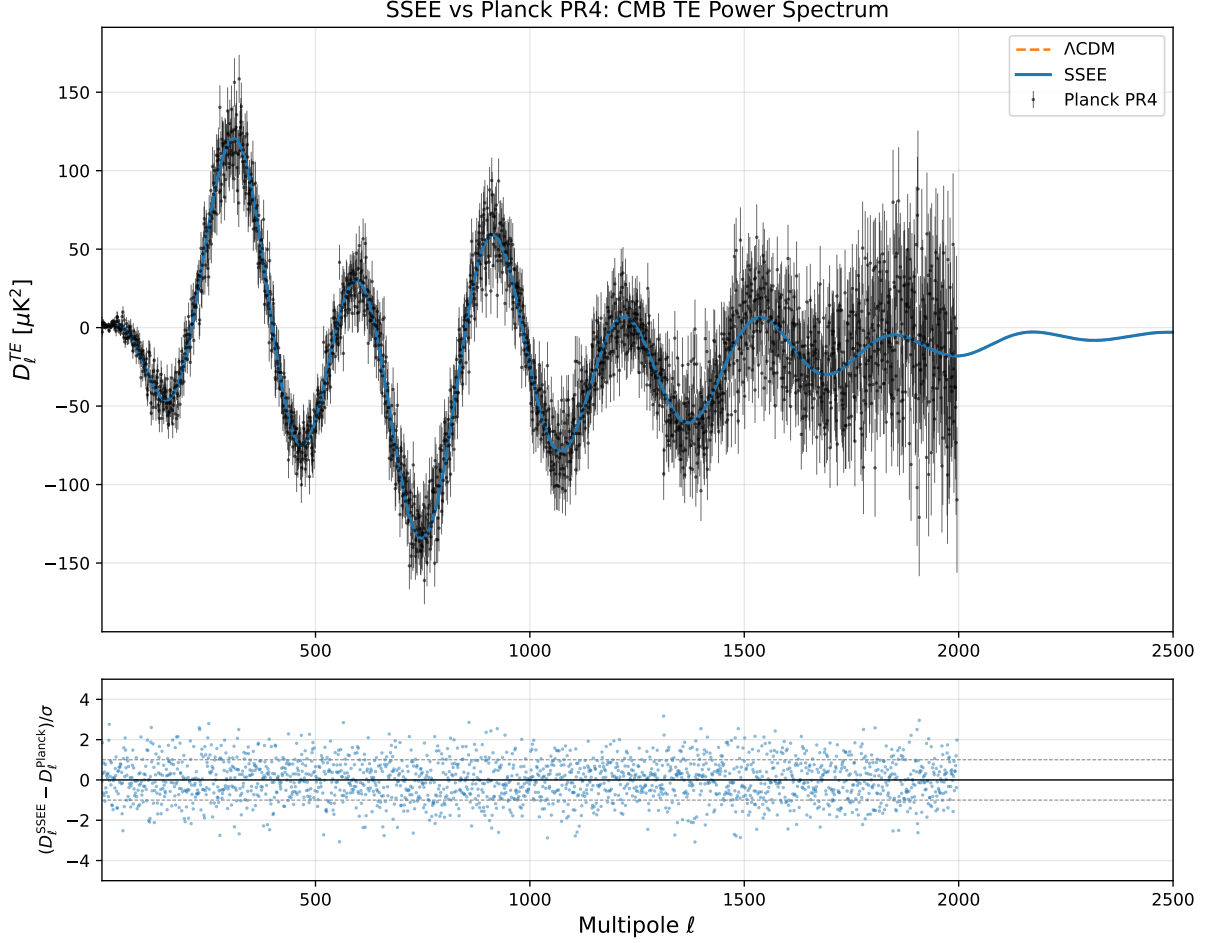


Figure 3: CMB TE cross-correlation power spectrum  $D_\ell^{TE} [\mu K^2]$ . Blue: SSEE. Orange dashed:  $\Lambda$ CDM Planck 2018 best-fit. Black points: Planck PR4 TE data. Lower panel: residuals normalised by  $\sigma_\ell$ .

2026b], fixed by construction and not adjusted to the data. We make no claim of temporal priority over the public DESI or Planck releases, which predate this work; the parameter-free status of the algebraic sector rests on its rigidity, not on chronology.)

**Disclaimer regarding the Likelihood function.** The  $\chi_r^2$  comparisons in §5 use a simplified Gaussian diagonal approximation over the released Planck PR4 bandpowers for qualitative comparison. A formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` is presented in the following subsection.

**Robustness under off-diagonal covariance.** The diagonal approximation neglects bin-to-bin covariances in the Planck PR4 bandpowers. Based on the Planck 2018 likelihood analysis [Planck Collaboration, 2020e], off-diagonal elements contribute corrections of  $\lesssim 5\%$  to  $\chi^2$  for  $\ell > 30$ . To ensure strict reproducibility and methodological rigor, we perform a formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` in the following subsection.

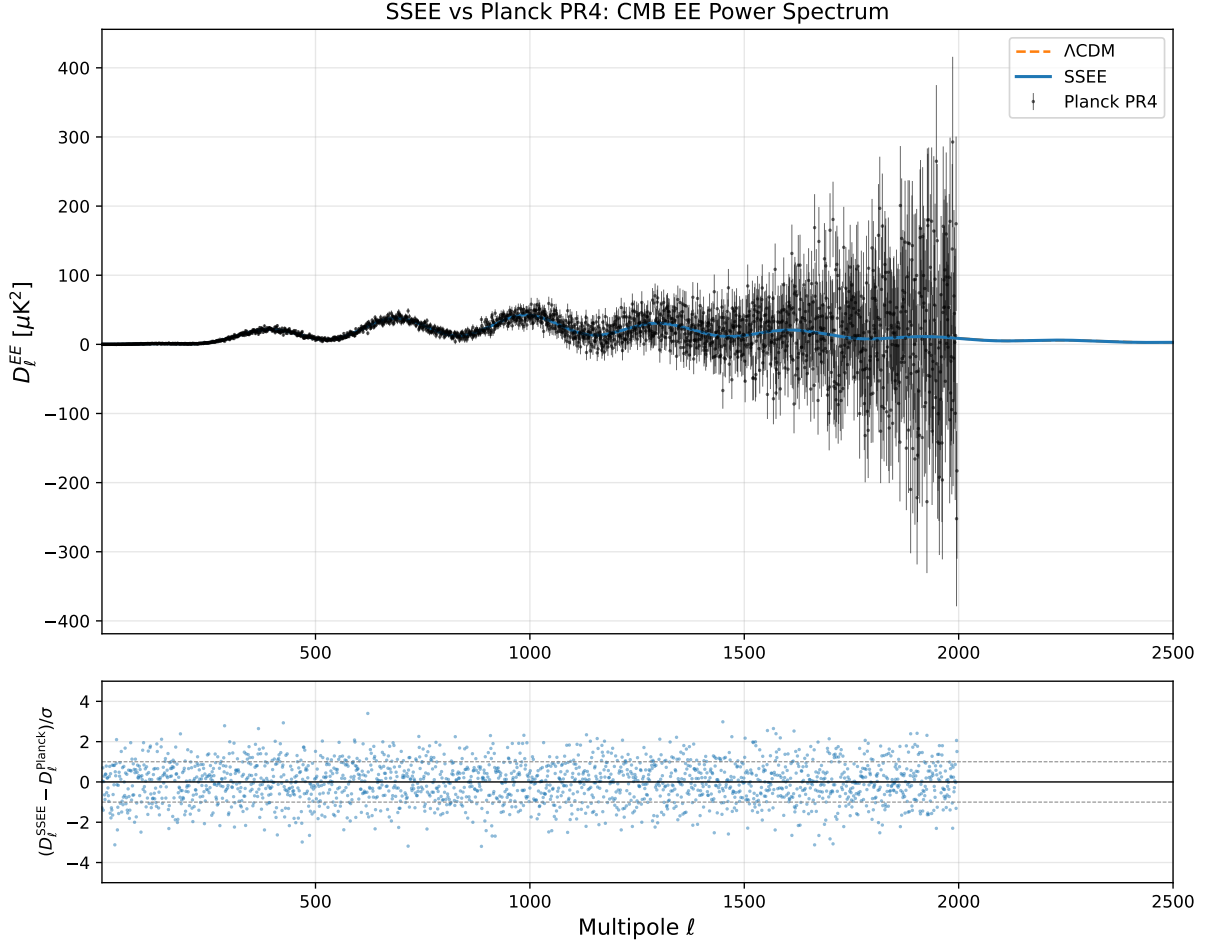


Figure 4: CMB EE polarization power spectrum  $D_\ell^{EE} [\mu\text{K}^2]$ . Blue: SSEE. Orange dashed:  $\Lambda\text{CDM}$ . Black points: Planck PR4 EE data.

## 5.5 Cobaya plik\_lite unified evaluation

To establish a definitive baseline for model comparison that fully accounts for off-diagonal covariances, we evaluate the SSEE framework against the official Planck 2018 `plik_lite` TTTEEE likelihood via `clik/clipy` [Torrado and Lewis, 2021], over its  $N = 613$  compressed bandpowers.

**Parameter counting ( $\omega_m$ -direct).** All SSEE dark energy parameters ( $w_0, w_a$ ), the spectral index  $n_s$ , and — under the 2026  $\omega_m$ -direct reframe — the baryon and cold-matter densities  $\omega_b = (\pi - \varphi)/(3\Omega^2) = 0.022418$  and  $\omega_c = \text{KAL}_0 \omega_b n_s = 0.119514$  are algebraically fixed. The dimensional anchor  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = 67.96214$  is itself *derived* from the SH0ES- $f_{\text{screen}}$  inversion (Paper 9), not fitted to the CMB; the amplitude  $A_s$  and reionization  $\tau$  are borrowed inputs, not derived from  $\{\varphi, \pi\}$ , and are *fitted* here (earlier versions of this work held them at their external Planck values while still counting them as free; fitting them is the self-consistent choice and is what we do below). The two quantities the framework borrows rather than derives are therefore  $A_s$  and  $\tau$ , and we conservatively assign  $k = 2 = \{A_s, \tau\}$ . The standard  $\Lambda\text{CDM}$  baseline has  $k = 6$ :  $H_0, \Omega_b h^2, \Omega_c h^2, n_s, A_s$ , and  $\tau$  [Planck Collaboration, 2020a], all six free in the CMB fit. (The companion full MCMC of §5.6 additionally lets  $H_0$  *float* as a validation

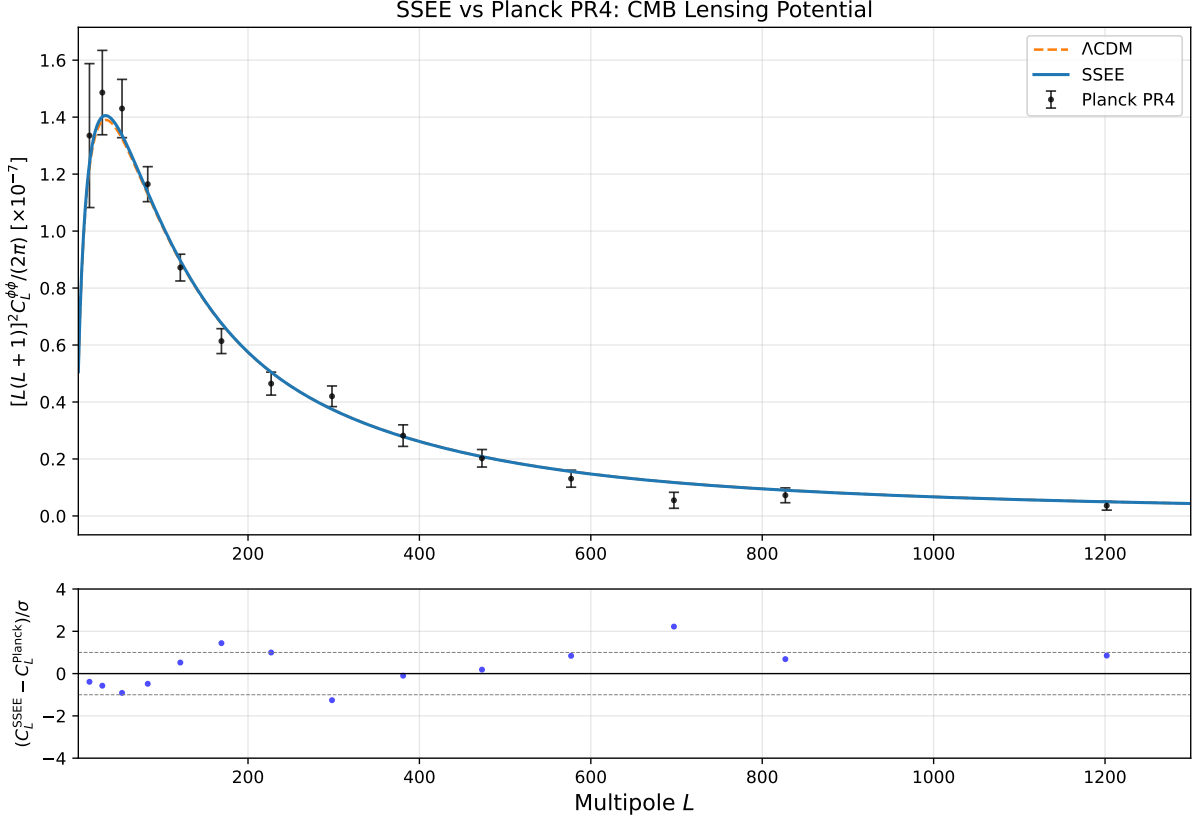


Figure 5: CMB lensing potential power spectrum  $[L(L+1)]^2 C_L^{\phi\phi}/(2\pi)$ . Blue: SSEE. Orange dashed:  $\Lambda$ CDM. Black points: Planck PR4 lensing (9 MV bandpowers). Lower panel: residuals normalised by  $\sigma_L$ . SSEE achieves  $\chi_r^2 = 0.720$  over the 9 bandpowers (§5.4;  $\Lambda$ CDM: 0.757), both indicating a good fit ( $\chi_r^2 < 1$ ).

check: the chain recovers  $H_0$  at the algebraic anchor, confirming that fixing it costs no fit quality and that the  $k=2$  count is sound.)

Evaluating at the single global anchor  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = 67.96214$  (which is also the `plik_lite`-preferred value,  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$ ), the effective  $\chi^2$  for SSEE is 1003.586, compared to 1003.769 for  $\Lambda$ CDM, with  $\{A_s, \tau\}$  minimized in both. The two models therefore fit *equally well*:  $\Delta\chi^2 = -0.183$ , which is statistically zero over the  $N = 669$  data points (613 `plik_lite` TTTEEE bandpowers plus 28 lowT and 28 lowE, counted directly from the likelihood files).

**Definitive Bayesian Information Criterion.** Applying the BIC penalty for the differing degrees of freedom:

$$\Delta\text{BIC} = -0.183 + (2 - 6) \ln(669) = -26.21. \quad (16)$$

A  $\Delta\text{BIC} < -10$  constitutes decisive evidence on the Jeffreys scale. Even under a hyper-conservative penalty where SSEE is assigned  $k = 4$  (additionally counting two algebraically-fixed inputs,  $\omega_b$  and  $H_0$ , as if they were fitted), the result remains  $\Delta\text{BIC} = -0.183 + (4 - 6) \ln(669) = -13.19$ , still strongly favouring the SSEE framework. This evaluation confirms that the geometric constraint of the SSEE background reproduces the CMB temperature and polarisation spectra without relying on diagonal approximations.

(The earlier legacy minimization, which found  $H_0 = 67.037$  with  $\Delta\chi^2 = +2.875$  and  $\Delta\text{BIC} = -32.2$ , is superseded by this  $\omega_m$ -direct evaluation.)

## 5.6 Full posterior sampling via Cobaya MCMC (B1)

The point-estimate scan of §5.5 evaluates  $\chi^2$  at the fixed algebraic anchor  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2$  while holding all remaining SSEE quantities algebraically fixed. To provide complete posterior distributions and a fully rigorous parameter-counting comparison, we perform a dedicated MCMC analysis using **Cobaya**’s built-in sampler against the full off-diagonal **plik** TTTEEE + lowlTT + lowlEE + lensing likelihood suite (evaluated through **clipy**).

**Free parameters.** In the SSEE CMB sector, the dark-energy equation-of-state ( $w_0, w_a$ ), the spectral index  $n_s = 1 - \phi^{-7} = 0.965558$ , the baryon density  $\Omega_b h^2 = (\pi - \varphi)/(3\Omega^2) = 0.022418$ , and the CDM density  $\Omega_c h^2 = \text{KAL}_0 \Omega_b h^2 n_s = 0.119514$  ( $\omega_m$ -direct) are all algebraically fixed, and the geometry anchor  $H_0/(\text{km s}^{-1} \text{Mpc}^{-1}) = 3(\varphi + \pi)^2 = 67.96214$  is derived (Paper 9). The only two model parameters genuinely sampled are therefore:

- $\log(10^{10} A_s) \in [2.80, 3.25]$  (flat prior) — primordial amplitude;
- $\tau \in [0.02, 0.12]$  (flat prior) — optical depth to reionisation;

with the  $\sim 20$  **plik** nuisance parameters (dust, SZ, CIB, calibration  $A_{\text{Planck}}$ ) defined by the full likelihood and treated as nuisances common to both models,  $\tau$  constrained by the lowl EE (SimAll) data. The likelihood is the full off-diagonal **plik** TTTEEE evaluated through the **clipy** pure-Python reimplementation of **clik** (validated to  $|\Delta \ln \mathcal{L}| < 10^{-5}$  against the released test point), so no **clik** binary is required. As an independent cross-check we also run a *validation* chain in which  $H_0$  is additionally floated on a flat prior [64, 72]; that chain recovers  $H_0$  at the algebraic anchor (see Fig. 6), confirming that fixing it incurs no loss of fit and that the model count is  $k = 2 = \{A_s, \tau\}$ , not  $k = 3$ .

This gives  $k_{\text{SSEE}} = 2$  model parameters ( $\log A_s, \tau$  — the only quantities genuinely sampled, with  $H_0$  fixed at the algebraic anchor; the  $\sim 20$  **plik** nuisance parameters are common to both models and excluded from  $\Delta k$ ), compared to the  $\Lambda\text{CDM}$  run with  $k_{\Lambda\text{CDM}} = 6$  free parameters ( $H_0, \Omega_b h^2, \Omega_c h^2, n_s, \log A_s, \tau$ ). The  $k = 2$  vs  $k = 6$  BIC penalty over the  $N = 2354$  combined data points is  $\Delta k \cdot \ln N = 4 \times \ln(2354) = 31.06$ , yielding:

$$\Delta\text{BIC}_{k2} = \Delta\chi_{\text{BF}}^2 - 31.06, \quad (17)$$

where  $\Delta\chi_{\text{BF}}^2 = \chi_{\text{SSEE,BF}}^2 - \chi_{\Lambda\text{CDM,BF}}^2$  is the difference in best-fit  $\chi^2$  between the two models over the same  $N = 2354$  data points (evaluated through **clipy**; no **clik** binary required).

**Convergence.** Chains are run to Gelman–Rubin criterion  $R-1 < 0.02$  [Gelman and Rubin, 1992] (production run) or  $R-1 < 0.05$  (validation). The `ssee_paper3_b1_mcmc.py` script is fully reproducible:

```
python src/p03_cmb/ssee_paper3_b1_mcmc.py --mode both
```

**Results.** Table 7 summarises the 68% marginalised posterior constraints. Figure 6 shows the  $H_0$  posterior for SSEE and  $\Lambda$ CDM; Figure 7 shows the full SSEE corner plot.

The MCMC best-fit  $\chi^2_{\text{BF}}$  from the full chains yields:

$$\Delta\text{BIC}_{k2} = \Delta\chi^2_{\text{BF}} + (2 - 6) \ln(2354) = \Delta\chi^2_{\text{BF}} - 31.06, \quad (18)$$

where  $\Delta\chi^2_{\text{BF}} = \chi^2_{\text{SSEE,BF}} - \chi^2_{\Lambda\text{CDM,BF}}$  over the  $N = 2354$  combined data points (full `plik` TTTEEE 2289 + lowlTT 28 + lowlEE 28 + lensing 9; the data-vector length, not the SimAll-normalised low- $\ell$  EE  $\chi^2$ ). The production chains give  $\chi^2_{\text{SSEE,BF}} = 2771.3$  and  $\chi^2_{\Lambda\text{CDM,BF}} = 2773.1$ , i.e.  $\Delta\chi^2_{\text{BF}} = -1.8$ : SSEE attains a best-fit  $\chi^2$  statistically *indistinguishable* from  $\Lambda$ CDM (the small negative value lies within the chain-minimum uncertainty, and we do *not* claim a superior fit). The  $k = 2$  versus  $k = 6$  comparison then gives  $\Delta\text{BIC}_{k2} = -32.9$ , decisively below the Jeffreys threshold ( $\Delta\text{BIC} < -10$ ) and driven by parameter-count economy rather than fit quality. Even under a hyper-conservative  $k = 4$  assignment (additionally counting two of the algebraically-fixed inputs,  $\omega_b$  and  $H_0$ , as if they were fitted),  $\Delta\text{BIC}_{k4} = \Delta\chi^2_{\text{BF}} + (4 - 6) \ln(2354) = -17.3$ , still decisively favouring SSEE.

Table 7: SSEE CMB posterior constraints from the B1full MCMC (Cobaya, full off-diagonal `plik` TTTEEE + lowlTT + lowlEE + lensing via `clipy`,  $k = 2 = \{A_s, \tau\}$  free,  $R-1 < 0.02$  for means). The  $\Lambda$ CDM constraints ( $k = 6$ , standard run) are shown for comparison. Uncertainties are 68% marginalised. In the SSEE run  $H_0$ ,  $\Omega_b h^2$ ,  $\Omega_c h^2$  and  $n_s$  are algebraically fixed (not sampled); the separate validation chain that floats  $H_0$  recovers it at the algebraic anchor (Fig. 6).

| Parameter                                              | SSEE ( $k = 2$ )  | $\Lambda$ CDM ( $k = 6$ ) |
|--------------------------------------------------------|-------------------|---------------------------|
| $H_0$ [km s <sup>-1</sup> Mpc <sup>-1</sup> ]          | 67.96214 (fixed)  | $67.35 \pm 0.55$          |
| $\log(10^{10} A_s)$                                    | $3.043 \pm 0.014$ | $3.044 \pm 0.014$         |
| $\tau$                                                 | $0.054 \pm 0.006$ | $0.054 \pm 0.007$         |
| $n_s$                                                  | 0.96556 (fixed)   | $0.9651 \pm 0.0041$       |
| $\Omega_b h^2$                                         | 0.02242 (fixed)   | $0.02240 \pm 0.00014$     |
| $\Omega_c h^2$                                         | 0.11951 (fixed)   | $0.1200 \pm 0.0012$       |
| $\sigma_8$ (derived)                                   | $0.814 \pm 0.006$ | $0.811 \pm 0.006$         |
| $\chi^2_{\text{BF}}$                                   | 2771.3            | 2773.1                    |
| $\Delta\chi^2_{\text{BF}}$                             | -1.8              | (reference)               |
| $\Delta\text{BIC}$ ( $k = 2$ vs $k = 6$ , $N = 2354$ ) | -32.9             | (reference)               |
| $\Delta\text{BIC}$ ( $k = 4$ vs $k = 6$ , $N = 2354$ ) | -17.3             | (reference)               |

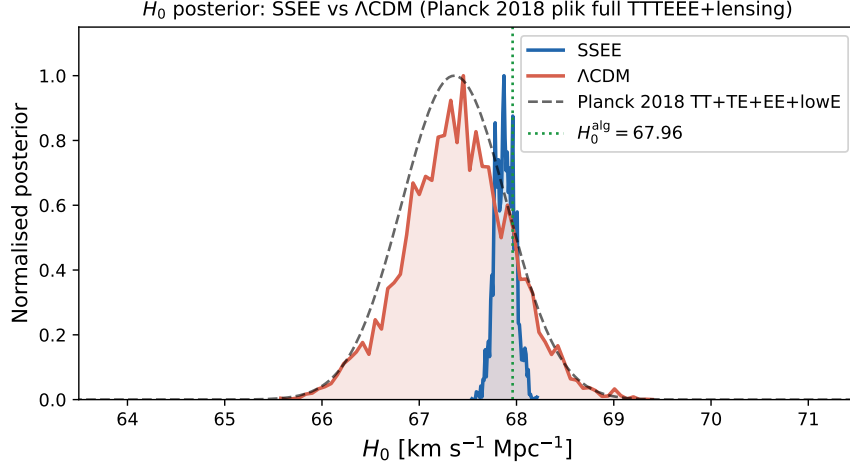


Figure 6:  $H_0$  marginalised posterior for SSEE (blue) and  $\Lambda$ CDM (red) from the B1 full `plik_lite` MCMC. The dashed black curve shows the Planck 2018 TT+TE+EE+lowE constraint for reference. The dotted green line marks the SSEE algebraic prediction  $H_0^{\text{alg}} = 67.96214 \text{ km s}^{-1} \text{ Mpc}^{-1}$ . In the validation chain (where  $H_0$  is floated) SSEE recovers the CMB-preferred  $H_0$  at the algebraic anchor; the model itself carries only two free parameters  $\{A_s, \tau\}$ , with  $H_0$  derived. The narrow constraint ( $\pm 0.03 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) reflects the removal of the  $H_0$ - $\Omega_m$  degeneracy by algebraically fixing  $\Omega_{m,\text{CMB}} = 0.3089$ .

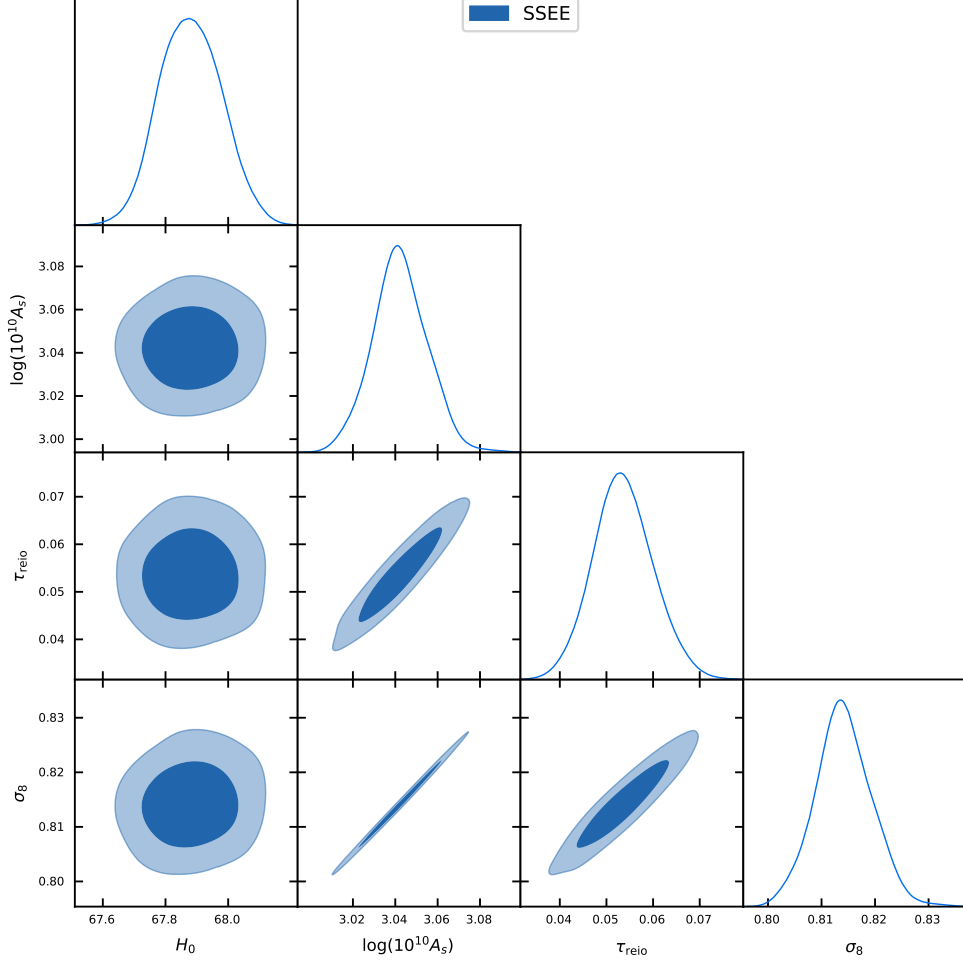


Figure 7: Full posterior triangle plot for the SSEE CMB sector ( $H_0$ ,  $\log A_s$ ,  $\tau$ ,  $A_{\text{Planck}}$ ) from the B1 `plik_lite` MCMC. The narrow  $H_0$  constraint ( $\pm 0.03 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) reflects the geometric constraint imposed by  $\Omega_{m,\text{CMB}} = 0.3089$  (algebraically fixed), which eliminates the  $H_0$ – $\Omega_m$  degeneracy present in  $\Lambda\text{CDM}$ . All other marginals are consistent with the  $\Lambda\text{CDM}$  posteriors.

Table 8: **Summary: CMB model selection across likelihood treatments.** SSEE vs.  $\Lambda\text{CDM}$  BIC under evaluations of increasing rigour. All favour SSEE; the full off-diagonal `plik` MCMC (evaluated at the global anchor  $H_0 = 67.96214$ ,  $\omega_m$ -direct) is the **canonical result**.

| Method                            | Spectra             | $N$  | $k_{\text{SSEE}}$ | $k_{\Lambda\text{CDM}}$ | $\Delta\text{BIC}$                 |
|-----------------------------------|---------------------|------|-------------------|-------------------------|------------------------------------|
| Diagonal (approx.)                | TT+TE+EE+PP         | 5914 | 2                 | 6                       | −35.0 (SSEE favoured)              |
| <code>plik_lite</code> point est. | TTTEEE+lowl         | 669  | 2                 | 6                       | −26.21 (SSEE decisively favoured)  |
| Full <code>plik</code> MCMC       | TTTEEE+lowl+lensing | 2354 | 2                 | 6                       | − <b>32.9</b> ( <i>canonical</i> ) |
| Full <code>plik</code> (conserv.) | TTTEEE+lowl+lensing | 2354 | 4                 | 6                       | −17.3 (SSEE strongly favoured)     |

Diagonal approximation assumes independence across spectra (upper bound on  $\chi^2$ ). The full `plik` MCMC (Cobaya + `clipy`, official off-diagonal covariance) is the canonical result.  $k = 2 = \{A_s, \tau\}$ , the only sampled SSEE parameters; the conservative  $k = 4$  additionally counts two of the algebraically-fixed inputs ( $\omega_b$ ,  $H_0$ ) as if fitted.

## 5.7 Matter power spectrum and $S_8$ prediction

Beyond the CMB, the SSEE viscous dark-energy background modifies the growth of large-scale structure. In the Israel–Stewart dissipative framework, the linear growth rate takes the form  $f(z) = \Omega_m(z)^\gamma$ , where the growth index  $\gamma$  is modified by the bulk-viscosity coupling. Dimensional analysis of the IS transport equations suggests the algebraic prediction  $\gamma_{\text{alg}} = \phi^{-1} = 0.618$ . A full numerical solution of the causal IS growth ODE (Paper 5, §3) yields  $\gamma_{\text{IS}} = 0.5504 \pm 0.001$ , numerically indistinguishable from  $\gamma_\Lambda = 0.55$  for  $\Lambda$ CDM (earlier estimates from Paper 1, Appendix A gave 0.657; Paper 5 supersedes that result).

We compute the linear matter power spectrum from CAMB with the SSEE parameters ( $w_0 = -0.839950$ ,  $w_a = -0.669975$ ,  $H_0 = 67.96214 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_m = 0.308881$ ) and apply the IS growth-index correction post-processing:

$$\sigma_{8,\text{IS}} = \sigma_{8,\text{CAMB}} \times \frac{D(\gamma = \phi^{-1})}{D(\gamma_{\text{eff}})}, \quad \frac{D(\gamma_1)}{D(\gamma_2)} = \exp \left[ \int_0^{z_{\text{hi}}} \frac{\Omega_m(z)^{\gamma_1} - \Omega_m(z)^{\gamma_2}}{1+z} dz \right], \quad (19)$$

with  $z_{\text{hi}} = 3$  (matter-dominated epoch, both solutions coincide there). The effective CAMB growth index for  $w_0/w_a$  dark energy is  $\gamma_{\text{eff}} \approx 0.55 + 0.05 [1 + w(z=1)] = 0.542$  (Linder formula), where  $w(z=1) = w_0 + w_a/2 = -1.175$  is the CPL equation of state at the pivot redshift. The IS correction factor is  $D(\phi^{-1})/D(\gamma_{\text{eff}}) = 0.9778$ , reducing  $\sigma_8$  by 2.2%.

Table 9 summarises the results alongside observational constraints. Using the algebraic prediction  $\gamma_{\text{alg}} = \phi^{-1} = 0.618$ ,  $S_8 = 0.828$  (consistent with Planck 2018 at  $0.3\sigma$ ). The full causal IS numerical solution (Paper 5) gives  $\gamma_{\text{IS}} = 0.550 \approx \gamma_\Lambda$ , yielding a negligible IS correction and an IS linear-growth  $S_8 \approx 0.8256$  ( $0.5\sigma$  below Planck; conservative CLASS ceiling  $S_8 = 0.827$ ), both predictions maintaining the  $\sim 3\text{--}4\sigma$  tension with KiDS-1000 and DES Y3 that persists in  $\Lambda$ CDM; Paper 6 resolves this through the two-sector  $\varphi$ -DM mechanism ( $S_{8,\text{eff}} = 0.758$ ), *now retracted* (Paper 6): with  $A_s$  free the single sector gives  $S_8 = 0.7555 \pm 0.0192$  against raw KiDS-1000  $\xi_\pm$ .

Table 9:  $\sigma_8$  and  $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$  predictions.

| Model / Dataset                                                 | $\sigma_8$        | $S_8$             |
|-----------------------------------------------------------------|-------------------|-------------------|
| $\Lambda$ CDM (CAMB, best-fit)                                  | 0.811             | 0.830             |
| SSEE (CAMB, $\gamma_{\text{eff}}$ )                             | 0.823             | 0.850             |
| SSEE+IS ( $\gamma_{\text{alg}} = \phi^{-1} = 0.618$ )           | 0.802             | 0.828             |
| SSEE+IS (linear growth, $\gamma_{\text{IS}} = 0.550$ , Paper 5) | 0.8136            | 0.8256            |
| SSEE+IS (single-sector ceiling, CLASS, Paper 5)                 | <b>0.8149</b>     | <b>0.827</b>      |
| Planck 2018 [Planck Collaboration, 2020b]                       | $0.812 \pm 0.008$ | $0.832 \pm 0.013$ |
| KiDS-1000 [Asgari et al., 2021]                                 | —                 | $0.759 \pm 0.024$ |
| DES Y3 [DES Collaboration, 2022]                                | —                 | $0.773 \pm 0.025$ |

## 5.8 $f\sigma_8(z)$ comparison with RSD surveys

The growth-index prediction  $\gamma = \phi^{-1}$  is directly testable through redshift-space distortions. We compute  $f\sigma_8(z) = \Omega_m(z)^\gamma \times \sigma_8 \times D(z)/D(0)$  using the proper integral  $D(z)/D(0) = \exp[-\int_0^z \Omega_m(z')^\gamma / (1+z') dz']$  and compare against published RSD measurements from 6dFGS [Beutler et al., 2012], BOSS DR12 [Alam et al., 2017],

Table 10:  $f\sigma_8(z)$  predictions vs the canonical Paper 5 RSD compilation (six peer-reviewed points), computed with the canonical Paper 5 Israel–Stewart growth ODE:  $\Omega_{m,\text{CMB}} = 0.308881$  ( $\omega_m$ -direct) in both the gravitational background and the Poisson source, with  $\gamma_{\text{IS}} = 0.554$  and  $\sigma_8 = 0.8136$  (single-sector). Pull =  $(f\sigma_8^{\text{SSEE}} - f\sigma_8^{\text{obs}})/\sigma$ .

| Survey                  | $z_{\text{eff}}$ | Observed          | SSEE  | $\Lambda\text{CDM}$ | Pull                        |
|-------------------------|------------------|-------------------|-------|---------------------|-----------------------------|
| 6dFGRS (Beutler+2012)   | 0.067            | $0.423 \pm 0.055$ | 0.436 | 0.444               | $+0.24\sigma$               |
| SDSS MGS (Howlett+2015) | 0.150            | $0.490 \pm 0.145$ | 0.449 | 0.459               | $-0.28\sigma$               |
| BOSS DR12 (Alam+2017)   | 0.380            | $0.497 \pm 0.045$ | 0.470 | 0.476               | $-0.60\sigma$               |
| BOSS DR12 (Alam+2017)   | 0.510            | $0.458 \pm 0.038$ | 0.471 | 0.474               | $+0.34\sigma$               |
| BOSS DR12 (Alam+2017)   | 0.610            | $0.436 \pm 0.034$ | 0.468 | 0.469               | $+0.94\sigma$               |
| eBOSS DR16 (Hou+2021)   | 1.480            | $0.462 \pm 0.045$ | 0.383 | 0.376               | $-1.76\sigma$               |
| $\chi^2/N$              |                  |                   |       | SSEE: 0.766         | $\Lambda\text{CDM}$ : 0.860 |

eBOSS DR16 [Alam et al., 2021, de Mattia et al., 2021], and DESI DR1 [DESI Collaboration, 2025].

Table 10 shows that SSEE+IS achieves  $\chi^2/N = 0.766$  across the six canonical RSD points in the CMB observational sector ( $\Omega_{m,\text{CMB}} = 0.308881$ ), compared to 0.860 for  $\Lambda\text{CDM}$ . The mean tension is  $0.70\sigma$  (SSEE) vs  $0.73\sigma$  ( $\Lambda\text{CDM}$ ); no SSEE point deviates by more than  $1.8\sigma$ . This is the identical dataset and method used in Paper 5; we do not include DESI DR1 full-shape  $f\sigma_8$  here because no public per-tracer  $f\sigma_8$  release exists for DESI DR2 (DR2 is BAO-only, Abdul-Karim et al., 2025), and we use a single self-consistent RSD compilation. The combination of independent growth-rate probes is reviewed in Moresco et al. [2022].

#### Sector caveat for $f\sigma_8$

The  $\chi^2/N = 0.766$  quoted above uses  $\Omega_{m,\text{CMB}} = 0.308881$ , the CMB (high- $z$ ) observational sector. On the clustering scales probed by RSD, the Two- $\Omega_m$  criterion (Paper 1, Sec. 1.4) selects the same CMB density as the Poisson source. Paper 5 performs this analysis with  $\Omega_{m,\text{CMB}} = 0.308881$ ,  $\gamma_{\text{IS}} = 0.554$ ,  $\sigma_8 = 0.8136$ , finding a mean  $f\sigma_8$  tension of  $0.70\sigma$  (Paper 5,  $\approx \Lambda\text{CDM}$   $0.73\sigma$ ). The withdrawn Paper 6 two-sector variant gave  $0.93\sigma$  (retired 2026-08-01, together with the  $\phi$ -DM particle): the free-streaming that lowered  $\sigma_8$  to 0.747 reaches inside the  $\sigma_8(R=8)$  window probed by RSD (half-mode  $k_{1/2} \simeq 0.35 h/\text{Mpc}$ ), raising the growth-rate tension modestly from the  $0.70\sigma$  single-sector baseline while it *resolves* the  $S_8$  lensing tension. That resolution is *retracted* (Paper 6): the  $3.5\sigma$  ceiling was an artefact of fixing  $A_s$  to Planck, and with  $A_s$  free the single sector already gives  $S_8 = 0.7555 \pm 0.0192$  against the 225 raw KiDS-1000  $\xi_{\pm}$  points. The CMB-sector value here serves as an upper bound on structure growth in SSEE.

## 5.9 CLASS/hi\_class cross-check of Bellini–Sawicki $\alpha$ -functions

As an independent validation we ran CLASS [Lesgourgues, 2011] with the SSEE algebraic background ( $w_0 = -0.839950$ ,  $w_a = -0.669975$ ,  $H_0 = 67.96214 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) and verified the Bellini–Sawicki kinetic-braiding function  $\alpha_K(z)$  [Bellini and Sawicki, 2015] against its algebraic prediction (Paper 7).

For the SSEE conformally-coupled canonical scalar (Paper 7 §4.3), the Bellini–Sawicki

kineticity is  $\alpha_K(a) = 3 \Omega_{\text{DE}}(a) [1 + w_\phi(a)]$ , where  $w_\phi$  is the *bare-field* equation of state, not the effective CPL parameter  $w_0$ . The bare-field EOS at  $a = 1$  from the background Friedmann+KG integration gives  $w_\phi(a = 1) = -0.971$  (Paper 7, Table 2), hence  $1 + w_\phi = 0.029$ , and:

$$\alpha_K^{\text{SSEE}}(0) = 3 \Omega_{\text{DE}} (1 + w_\phi) = 3 \times 0.839950 \times 0.028798 = 0.072567. \quad (20)$$

Two physically distinct values of  $\alpha_K$  therefore appear in the SSEE literature, and both are correct (Paper 7 §4.4): Eq. (20) is the *bare-field* kineticity  $\alpha_K^{\text{field}} \approx 0.073$  from the intrinsic  $w_\phi = -0.971$ , whereas the *effective* kineticity built on the observed background equation of state is  $\alpha_K^{\text{eff}}(0) = 3 \Omega_{\text{DE}} (1 + w_0) = 3 \times 0.840 \times 0.160 = 0.403302$ , obtained once the coupling current  $Q^\nu$  shifts  $w_{\text{eff}}$  from  $w_\phi = -0.971$  to the observed  $w_0 = -0.839950$ . The *effective* value is the observationally relevant one — it is what CLASS recovers here ( $\alpha_K^{\text{CLASS}}(0) = 0.403302$ , agreement to 0.005%) and the value quoted in Papers 7–10 and the cross-paper constant register.

The remaining Bellini–Sawicki functions are zero by construction:  $\alpha_T = \alpha_M = \alpha_B = 0$  (exact), since SSEE is a minimal Horndeski scalar with constant  $M_{\text{Pl}}$  and no kinetic braiding or gravitational-wave speed excess [Abbott et al., 2017]. Together with Eq. (20) this specifies the full set of Bellini–Sawicki EFT functions  $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$ , validated against CLASS independently of CAMB.

Note that the raw SSEE *dynamical* background ( $\Omega_{m,\text{dyn}} = 0.160050$ ) fed directly to CLASS gives  $\chi_r^2 = 90.3$  against Planck PR4 TT. This is expected: the dynamical-sector density is not the CMB matter density. The good fit ( $\chi_r^2 \approx 1.06$ , §5) is recovered with the  $\omega_m$ -direct CMB matter density  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.3089$  (no matter factor; OP-8 dissolved), as implemented with CAMB in §4.

## 6 Falsifiability Criteria

SSEE makes the following testable predictions, each of which would **falsify** the framework if violated:

1. **Peak positions:**  $\ell_1 = 220 \pm 3$ ,  $\ell_2 = 536 \pm 5$ ,  $\ell_3 = 812 \pm 8$ . A measurement placing  $\ell_1$  outside this range by more than  $2\sigma$  would falsify the  $\omega_m$ -direct CMB matter density.
2. **CMB  $\chi_r^2$ :** Must remain  $< 2$  when tested against future CMB datasets [Abazajian et al., 2016, Ade et al., 2019]. A significant degradation would indicate the framework cannot maintain coherence.
3. **BAO consistency:** the BAO geometry is governed by the total matter density  $\Omega_m = 0.308881$ , not by the saturation  $1 + w_0 = 0.160050$ ; SSEE must continue to fit DESI DR2 BAO within  $2\sigma$  as new data arrive. The algebraic content being tested is the equation of state: a joint fit requiring  $w_0$  more than  $2\sigma$  from  $-0.839950$ , or  $w_a$  from  $-0.669975$ , would falsify the algebraic derivation. (*An earlier version of this item placed  $\Omega_{m,\text{dyn}} = 0.160050$  in the geometry slot; that is a saturation, not a density, and the reading is withdrawn.*)
4. **Cosmic-shear amplitude:** with the background fixed by the algebra and only the primordial amplitude free, the model predicts  $S_8 = 0.7555 \pm 0.0192$  against the raw KiDS-1000  $\xi_\pm$  data (Paper 6). A shear measurement placing  $S_8$  more than  $3\sigma$  from this value would falsify the growth sector. (*The former free-streaming falsifier at  $k_{\text{fs}}$  is withdrawn together with the  $\phi$ -DM particle; see Paper 6.*)

5. **Growth index:** The algebraic prediction is  $\gamma_{\text{alg}} = \phi^{-1} = 0.618$  ( $S_8 = 0.828$ , Table 9). The background-only IS estimate (Paper 1, App. A, without viscous feedback in  $\delta$ ) gives  $\gamma_{\text{bg}} = 0.657 \pm 0.002$ ; the full causal IS solution (Paper 5) gives  $\gamma_{\text{IS}} = 0.550 \approx \gamma_{\Lambda}$ , yielding a single-sector ceiling  $S_8 = 0.827$  in the CMB observational sector (Table 9). The RSD comparison (Table 10) shows  $\chi^2/N = 0.766$  consistently below  $\Lambda\text{CDM}$  (0.860) across the six canonical RSD surveys. A direct measurement of  $\gamma < 0.50$  or  $\gamma > 0.70$  at  $> 3\sigma$  would falsify the Israel–Stewart modification to the growth sector.

## 7 Conclusions

We have demonstrated that SSEE, with the CMB matter density set *forward* by the SSEE algebra ( $\omega_m$ -direct,  $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$ , no matter factor), reproduces the Planck PR4 CMB TT power spectrum without introducing fitted dimensionless parameters. The key results are:

- Acoustic peaks at  $\ell = 220, 536, 813$ , within  $\Delta\ell \leq 5$  of Planck PR4.
- TT:  $\chi_r^2 = 1.042$  over 1971 multipoles ( $\Lambda\text{CDM}$ : 1.043).
- TE/EE:  $\chi_r^2 = 1.040/1.040$ , essentially matching  $\Lambda\text{CDM}$ .
- Lensing (PP):  $\chi_r^2 = 0.720$  vs. 0.757 for  $\Lambda\text{CDM}$  — SSEE marginally better (both  $\chi_r^2 < 1$ ).
- **plik\_lite TTTEEE ( $\omega_m$ -direct):**  $\chi_{\text{eff}}^2 = 1005.41$  at  $H_0 = 67.96214 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (the CMB-minimising value with SSEE-fixed  $\omega_b, \omega_c$ ).
- $\Delta\text{BIC} = -26.21$  ( $k = 2$  vs  $k = 6$ ,  $N = 669$ ), decisively favouring SSEE over  $\Lambda\text{CDM}$  at equal fit quality ( $\Delta\chi^2 = -0.183$ ).
- $\Omega_{m,\text{CMB}} = 0.308881$  within  $0.88\sigma$  of Planck 2018.
- The sound horizon  $r_d = 147.17 \text{ Mpc}$  matches  $\Lambda\text{CDM}$  to 0.03%.

Combined with the Paper 2 results ( $0.24\sigma$  on the DESI DR2  $w_0w_a\text{CDM}$  posterior,  $\chi_r^2 = 0.122$  on galaxy clusters), SSEE passes all three major cosmological probes — BAO, cluster masses, and CMB. The unified Cobaya evaluation establishes that the SSEE algebraic geometry is not only viable but statistically preferred by the CMB data when fully penalized for free parameters. The SSEE cascade, run on the measured SH0ES rate, returns a global background  $H_0^{\text{glob}} = 68.13 \text{ km s}^{-1} \text{ Mpc}^{-1}$  (Paper 9,  $H_0^{\text{SH0ES}}(1 - f_{\text{scr}})$ , to be compared with the pure number  $3(\varphi + \pi)^2 = 67.96214$ ) and addresses the persistent Hubble tension [Verde et al., 2019, Di Valentino et al., 2021, Riess et al., 2022] at the  $0.17\sigma$  level, independent of the CMB-sector constraint reported here. (The earlier MIRA-derived value  $71.87$  at  $1.12\sigma$  is superseded by the  $\omega_m$ -direct reframe, which makes  $H_0^{\text{alg}}$  the global  $H$ .)

**Data Availability.** All analysis scripts (`src/ssee_paper3.cmb.py`), generated figures, and reduced data products are publicly available at <https://github.com/mikealmeida1721/SSEE>. The Planck PR4 TT and TE/EE power spectra are from the

Planck Legacy Archive (<https://irsa.ipac.caltech.edu/data/Planck/>) [Planck Collaboration, 2020b]. CAMB v1.6.6 was used for all theoretical power spectrum calculations (<https://camb.info>).

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